

Utilizing the Bradley-Terry Model to Measure the Competitive Balance of Men's Soccer Leagues

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Introduction

Sports have long played a significant social, cultural, and political role in many people's lives around the world, making sports leagues some of the biggest businesses in the world. Sports leagues want to have the best, most engaging matches to maximize viewership. This is why every sports league wants to be competitive, as not knowing the outcome of a given game makes for a more appealing product, thus increasing the league's revenue by drawing more viewers. (Fort, 2003). But, there is no standard way to measure the competitive balance of a sports league, as measures can range from simple standard deviation of wins to a complex model using advanced statistics (Alwell, 2020).

Quantifying the degree of parity within a sports league provides useful insights that guide structural decisions aimed at sustaining spectator interest, maximizing revenue potential, and promoting fair sporting competition across all teams (Zimbalist, 2002). To try and achieve this, I will utilize Bradley-Terry modeling's pairwise comparison techniques to evaluate competitive balance based on seasonal game outcomes spanning 2005 to 2023 for a selection of prominent professional men's soccer leagues internationally.

Pairwise comparisons are important tools that empower academic research across many different disciplines (Koczkodaj & Szybowski, 2015). At its core, a pairwise comparison simply involves examining any two ideas, variables, or products in relation to one another. By directly evaluating such items head-to-head, rather than in isolation, researchers gain additional insight in how one differs from another (Koczkodaj & Szybowski, 2015).

To attempt to solve this dilemma, I have created an index to measure the competitive balance of a given sports season given every head to head matchup in the league. This index utilizes a well known statistical model known as the Bradley-Terry Model, a probability model

for pairwise comparisons (Firth, 2005). To test my index, I will be looking specifically at men's professional soccer leagues from 2005-2023, in Europe and North America. Soccer's league structures are the most uniform and the vast amount of previous findings in soccer make it easy to determine the validity of my model thus making it the best choice for my goal.

Literature Review

Competitive Balance

To fully understand the scope of this paper we must first discuss the concept of competitive balance and its importance. A German economist named Tim Pawlowski who extensively analyzed the demand for international soccer broadcasts in America discusses this concept in a paper titled "The behavioural economics of competitive balance: Implications for league policy and championship management" (Budzinski & Pawlowski, 2014). This source specifically describes the behavioral economics aspect behind competitive balance and how leagues should implement certain policies. The paper concludes that perceived balance matters but only under specific conditions and restrictive interventions require strong justification. The paper begins by describing the standard theory of competitive balance which is that fans enjoy a league with more parity, leading them to be more willing to pay for the league. The next section justifies preserving balance using policy interventions such as revenue sharing, draft policy, and salary caps. Despite these claims, Budzinski & Pawlowski find that there has not been any evidence connecting competitive balance and attendance and revenue. However, it was found that fan attention and attendance were greater for things like relegation and title fights as opposed to overall balance of a league. When it comes to policy, the authors conclude that perceived balance matters but only under specific conditions and restrictive interventions require

strong justification (Budzinski & Pawlowski, 2014). This research is limited as it does not delve specifically into any sport and is only able to conclude with generalizations that may not apply to my research. However, this source provides implications of findings of the research and can possibly help suggest changes to the structure of some leagues to increase competitive balance.

Allen Sanderson, an economics professor at the University of Chicago, offers a different perspective on competitive balance with his paper “The many dimensions of competitive balance.” Sanderson describes the complexities of competitive balance in reference to a plethora of sports. He concludes by saying there is not a one-size-fits-all approach to the issue of differences in strengths, but there have been significant measures taken across sports. The paper discusses the ongoing debate on the influence of competitiveness on a leagues financial situation. The paper then discusses different methods of constraining competition, citing examples such as weight classes, draft systems, and age requirements (Sanderson, 2002). Despite their similar conjectures on the nature of competitive balance, Budzinski and Sanderson have some points of disagreement. Specifically, Budzinski seems to believe that competitive balance and revenue are not connected as there is no evidence to support this, while Sanderson believes that competitive balance and revenue are connected and cites how money can be used to constrain competitiveness. This further complicates the actions that leagues can take as there are opposing viewpoints on whether increasing competitive balance is really the optimal decision for a league.

Both sources on competitive balance agree that competitive balance is an incredibly important metric to all sports leagues and can provide lots of important information to the league, but there is no definite metric to measure competitive balance. However, it is also important to look at the current models used to measure competitive balance as they provide valuable context

for what my research is aiming to find and can serve as a source of comparison to determine the validity of the Bradley-Terry model itself.

Measuring Competitive Balance

The first measure of competitive balance comes from Francesca Basini and his fellow researchers' paper on “Assessing competitive balance in the English Premier League for over forty seasons using a stochastic block model.” The paper looks at a statistical network model based on an extension of the stochastic block model to analyze the competitive balance in the English Premier league over 40 seasons. The paper then goes season by season and analyzes the English Premier League and finds a major shift in balance around the 2000s. This shift took the league from relatively balanced to an essential two tier competition with the top teams being far better than the rest (Basini et al., 2023). This model is relatively complex compared to most metrics for competitive balance and can serve as a good point of comparison to my measure as it uses a statistical model that does not use pairwise comparison to make a competitive balance metric. Utilizing the results from this source will help establish the validity of pairwise comparison based models like the Bradley-Terry model to create a complex metric.

Another measure of competitive balance was created by Doria and Nalebuff, in their paper titled “Measuring competitive balance in sports.” This paper also looks at a different method of measuring the competitiveness of sports leagues except its key characteristic is that their measure is invariant to season length. The paper confirms that season length is an important factor when looking at a leagues competitiveness after analyzing the big four American sports leagues: NFL, NBA, MLB, and NHL. The paper concludes by attributing the least predictable leagues to the NFL and MLB due to the NFL's short season and the MLB's small dispersion of win probabilities (Doria & Nalebuff, 2023). In contrast to Basini's metric, this measure of

competitive balance is not derived from a different statistical model, however it does have a set equation. Doria and Nalebuff also thoroughly discuss the shortcomings of traditional measures of competitive balance such as a basic standard deviation of wins, which further reinforces the need for my metric.

The last paper on competitive balance I will examine comes from Kevin Alwell. His paper is called “Analyzing Competitive Balance in Professional Sport” and the paper highlights several measures of competitive balance and the variance in performance amongst the league. After discussing the importance of competitive balance Alwell hypothesizes that the NBA will be the least competitive among the top 5 sports leagues in the US. Alwell tests standard deviation, adjusted standard deviation, Gini coefficient, the Markov model for transitional probabilities to see the probability of a team going in and out of playoffs each year, and finally the Herfindahl-Hirschman Index (HHI) for measures of competitive balance (Alwell, 2020). The source is useful as it demonstrates many alternative measures to competitive balance that can be used as a point of comparison and further legitimize my model and use of pairwise comparisons. Alwell’s measures are all quite basic in comparison to the previous two metrics, but the Gini Coefficient and HHI are both similar to Basini’s measure in that they both are derived from more traditional statistical models. Through Alwell’s paper, one can see the possibility of overlap with other disciplines like economics as seen with his use of the Gini coefficient, a measure of wealth inequality. This establishes how far reaching the use of pairwise comparisons can be, possibly comparing entities like hospitals, schools, or even entire countries as the comparisons like sports leagues will be derived from many different factors (Elliot, 2010).

Bradley-Terry Model

To completely understand my competitive balance index, one must first understand the Bradley-Terry model. This model's use in sports is well articulated in the source "Dynamic bradley-terry modeling of sports tournaments" by Cattelan, Varin, and Firth, all applied statisticians. The paper defines the Bradley-Terry model itself as a probability model for the outcome of comparison between two things. The paper then applies the model in a dynamic paired comparison model to analyze the evolution of team abilities over the course of a sports tournament (Cattelan et al., 2012). This source adds relevant context to the overall purpose of my research by expanding upon a key aspect of the project in the Bradley-Terry model, which serves the same purpose in my paper as the stochastic block model in Basini's paper.

Gap

There is no specific model that can determine the competitiveness of a sports league, more specifically soccer. Many models have flaws such as being skewed for longer or shorter seasons or only taking into account final standings rather than looking at a season holistically (Doria & Nalebuff, 2023). The model I will be using is unique and something that can be applied to many different league's without it being skewed like other models. The model is also easy to understand and has the ability to be modified. Lastly, no paper looks at as many leagues and divisions of professional soccer as I do.

Method

Research Design

To accurately determine the extent to which competitive balance varies between soccer leagues, I will be using an evaluation analysis research method. In this method I first create the metric, using the Bradley-Terry model, for competitive balance. Then, I compare the value

obtained from that metric for each league to other leagues. This is the most common method used in the papers discussed above that also attempt to measure competitive balance such as by Doria and Nalebuff, who work in management at the NBA and Yale respectively (Doria & Nalebuff, 2023).

Method & Subjects of Study

A key part of my method is obtaining the data needed to run the Bradley-Terry model. All of this data is quantitative but applied to a more qualitative quality of a league in competitive balance. This data type best answers my research question as quantitative data is the best data to analyze the extent to which things differ and makes comparison between leagues much easier. In my method, I have no real subject of study besides the raw data sets I used, which is open source data from football-data.co.uk and kaggle.com/datasets/josephvm/major-league-soccer-dataset. (Football Data Staff, 2005; Mohr, 2022) These data sets compile every single match played in a season of a specific league and has each team's goals scored in separate columns allowing me to create a function that determines whether or not the home team has won (Appendix 2). The reason I selected these datasets is they were the most readily available and allowed for easy download and implementation.

Instruments

To apply this evaluation analysis method, I will utilize the Bradley-Terry model as the primary instrument to quantify competitive balance. As described in the literature review, the Bradley-Terry model is a probability model for the outcome of comparison between two things (Cattelan et al., 2012). Using the statistical model I will derive a metric for competitive balance by taking the variance team strength of every team in the league created by the model. To derive these team strengths, I essentially use the Bradley-Terry model backwards. Instead of using the

team strengths and home field advantage to calculate the probability of whether or not a team wins or loses a matchup (Atkinson, 1972). The Bradley-Terry model takes the given results of all the fixtures I input and back calculates the strength of each team to where the given outcomes are the ones of highest probability. I am using this instrument as it is an objective way to measure competitiveness of a soccer league and takes into account every single game played. In addition, the measure is not affected by season length or league structure, unlike other measures such as the Gini Coefficient and HHI(Doria & Nalebuff, 2023).

Procedure

To gather the majority of the match data for this I used the website football-data.co.uk. I downloaded the csv files for all leagues here as it had match by match data. How far the data went back varied from league to league with the top 5 European leagues going as far back as the early 90s but almost every other league going back as far as 2004. I then downloaded the data for all of my selected countries, dating back to 2005. Since the MLS had the most unique structure, I excluded playoff games and filled in the match data from [kaggle.com/datasets/josephvm/major-league-soccer-dataset](https://www.kaggle.com/datasets/josephvm/major-league-soccer-dataset).

To analyze all data, I used the programming language Julia on the platform Jupyter. After loading in the csv files as dataframes, each league was analyzed using my competitive balance metric. This came in the form of a new column which assigned a value which represents how competitive each season was (Appendix 1).

Instead of using the Bradley-Terry model for its traditional purpose of a probability model for the outcome of a head to head comparison, my competitive balance index uses the team strength assigned to each team following the analysis of each match and looks at the variance of this throughout the league. As soccer has wins, losses, and draws instead of only 2

outcomes, I added a column to each dataframe that asked if the home team won or drew rather than just won. I then ran the competitive balance model using both home team win and home team win or draw and averaged the strengths to get my result. Finally, after collecting all the competitive balance values for each and every season of each league I plotted the leagues using various filters. This plotting was also done through Julia and each different graph has a key as to what is being shown (Breloff, 2023). Once all the graphs and values were found, I began to compare the leagues to each other, plotting them on the same graph.

Results

First Divisions

To gather these results, I imported data for over 100,000 total soccer games from 2004 - 2023. Each season of each league was evaluated for competitive balance using the Bradley-Terry model and assigned a competitive balance index. Notably, this competitive balance index utilizes variance, and thus a lower score indicates a more competitive league. However, measures like this are meaningless without context so I will be viewing the results through different perspectives.

The first graph I will look at is of the top league in each of the ten countries I included in my study.

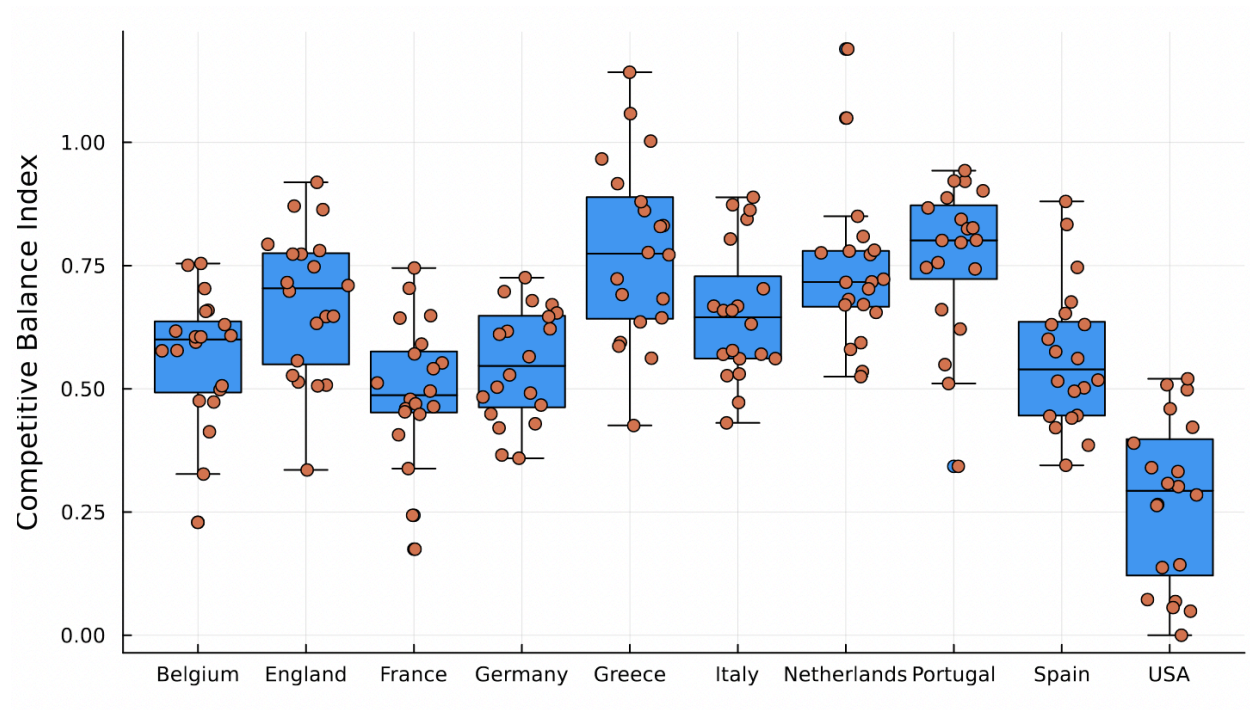


Figure 1: Competitive balance of first division soccer leagues organized by country from 2005-2023

This graph plots each individual season for each of the top divisions in their respective countries from 2004-2023. These individual values are superimposed on a box and whisker plot for each league displaying the mean, range, first quartile, and third quartile.

These results demonstrate that the most competitive league is the MLS (USA) as it has the lowest mean competitive balance index of the 10 countries I analyzed. Most other leagues have a relatively similar mean index, ranging anywhere from 0.5 to 0.7. The Portuguese and Greek first divisions seem to have the least competitive leagues on average with a value of 0.75. Out of the top five European leagues (Spain, Germany, England, Italy, France) it can be seen that France is the most competitive by a fair margin hovering at about 0.5. Generally the leagues

in smaller European countries are dominated by a few elite teams in the large cities which leads to higher competitive balance index.

Second Divisions

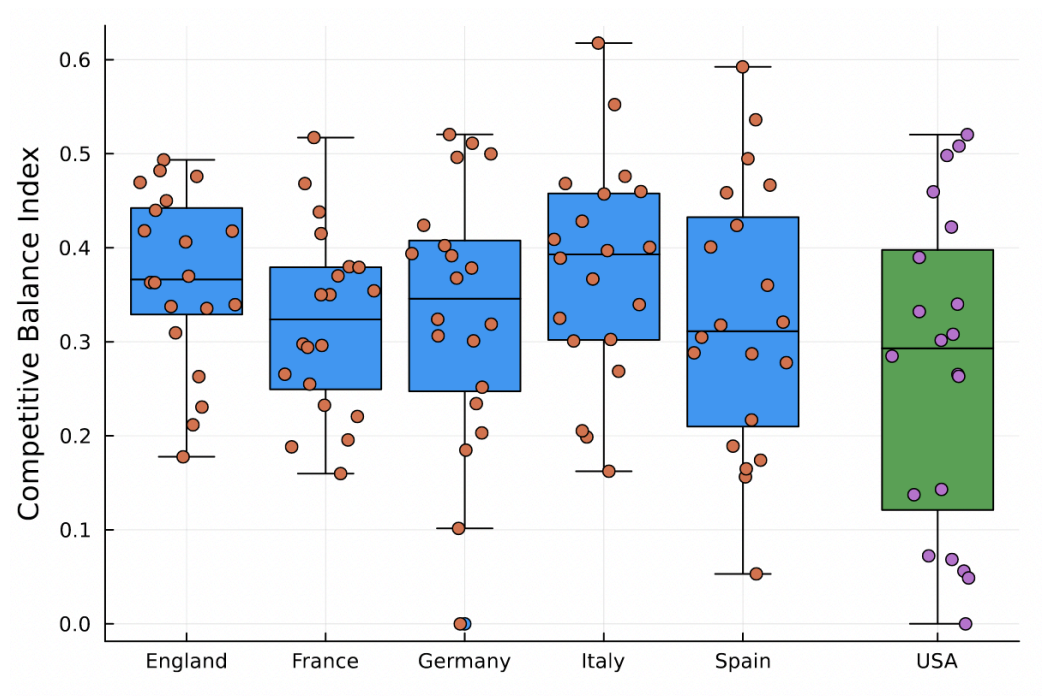


Figure 2: Competitive balance of second division soccer leagues organized by country

This second graph concerns the belief of many that the MLS to be a “second rate” league and thus compare it to lower tier leagues in top nations. To test this, I plotted the competitive balance of the MLS against the 2nd divisions in the top 5 European footballing leagues’ countries. As seen, the MLS is still slightly more competitive although they are all in a similar range when it comes to competitive balance. One thing to note is that the variance in competitive balance season to season seems to be highest in the MLS. In terms of competitive balance it

seems like the MLS is more comparable to second divisions in European soccer than first divisions, but the reason remains unknown.

England

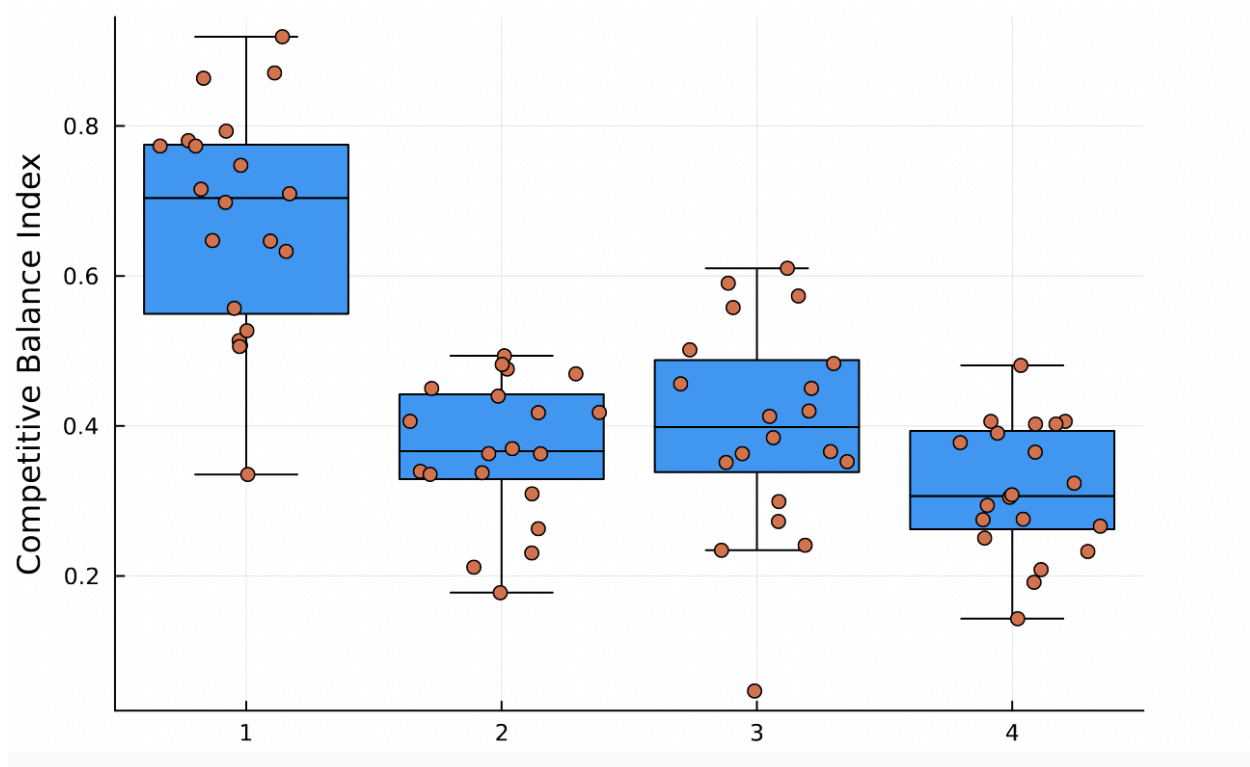


Figure 3: Competitive balance of professional English soccer leagues organized by division

To fully grasp the difference between first and second divisions, I will be taking a deeper dive into each of the top five European league's countries. The plot above illustrates the competitive index for the top 4 divisions of English soccer from 2005 to the present. As shown in Figure 3, the first division is significantly less competitive than the lower three. The lower divisions are all close in competitive balance with the 2nd division being surprisingly more

competitive than the third. All of these leagues, however, are larger than the first division (in amount of teams) and still more competitive.

Germany

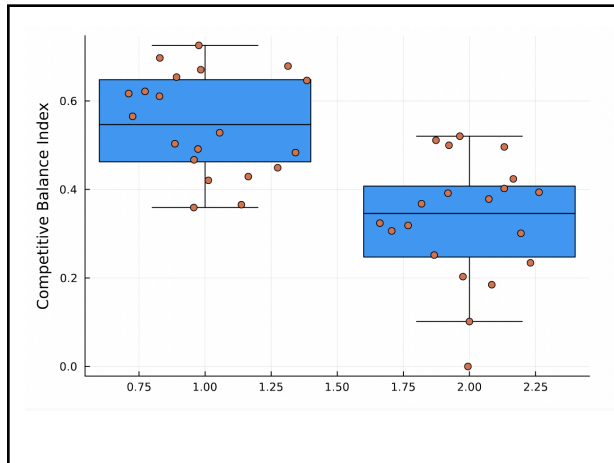


Figure 4: Competitive balance of professional German soccer leagues organized by division

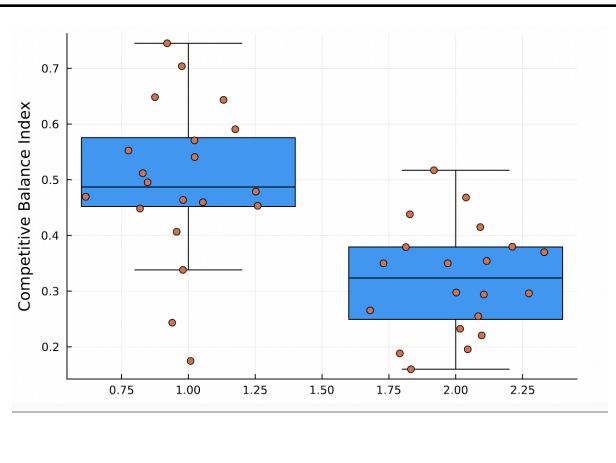


Figure 5: Competitive balance of professional French soccer leagues organized by division

Germany and France both demonstrate the same trend as England, as shown in the graph. with the Figures 4 and 5 above. The second division of Germany had a competitive balance index around .35 which is lower than the lowest competitive index from any first division season. Even with France being the most competitively balanced league in the top 5 European league's its second division is still more competitively balanced.

Spain and Italy

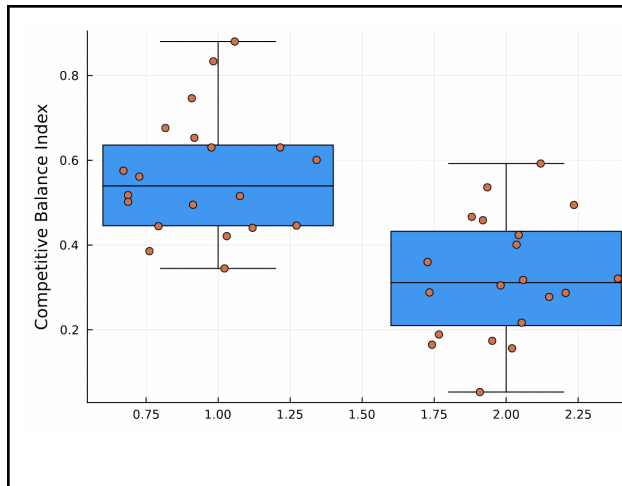


Figure 6: Competitive balance of professional Spanish soccer leagues organized by division

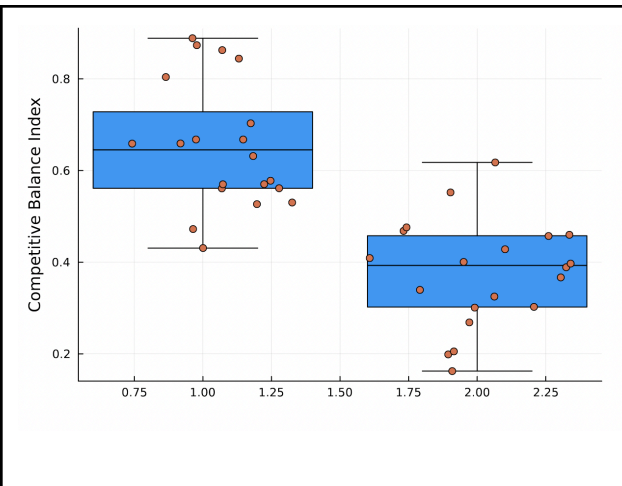


Figure 7: Competitive balance of professional Italian soccer leagues organized by division

The two graphs above further reinforce what the previous graphs also stated. In both countries the second division is significantly more competitive with an index of around .3, around half of the first divisions in their respective countries. With this, it is clear that all of the top five European leagues follow a similar trend in this with the second division being about twice as competitively balanced as the first.

Discussion

Validity of the Model

Before analyzing any results, I must determine the validity of the Bradley-Terry model as a measure of competitive balance.

There have been numerous studies conducted on the validity of the Bradley-Terry model as a measure of competitive balance. One of the such papers on this topic was "Measuring

Competitive Balance in Professional Team Sports Using the Herfindahl-Hirschman Index" by Rodney Fort and James Quirk (1995), who applied the Bradley-Terry model to Major League Baseball data and found it to be a reliable measure of competitive balance. They argued that the model was able to capture the dynamics of competitive balance in a league over time, as well as the impact of factors such as player salaries and team revenues on competitive balance (Fort & Quirk, 1995).

This same process can be applied to my results by comparing parts of them to the competitive balance measurements yielded by papers such as Basini et al. and Criado et al.

Basini et al. in particular looks at just the English Premier League over a longer period of time than me and utilizes the stochastic block model to determine competitive balance. Just as I identified, the 2016-17 season was one of the least competitive on the stretch of 2005-present. My results also seem to match the paper's in that the season's my model determined to be relatively competitive were also deemed the same by the stochastic block model (Basini et al, 2023).

But this may be just the English Premier League. To determine that my model's accuracy transfers between league's I can look at Criado et al. in which the researchers analyze the competitive balance of the Italian, German, Spanish, and English first divisions during the 2011-12 season and the 2012-13 season. Criado's use of a competitiveness graph, more commonly known as an ad hoc graph, yielded the exact same results as my Bradley-Terry model, recognizing the Premier League as the least competitive in 2012-13 followed by the Bundesliga and La Liga (Criado et al., 2013). This level of accuracy was consistent across the 2011-12 season as well.

Limitations

After comparing my findings to previous scholarly research, it is clear that the Bradley-Terry model accurately measures competitive balance. However, there are some limitations to my analysis. Firstly, I was only able to analyze the leagues with openly available public data, thus certain league's formats may not work as well with my model. In addition, the Bradley-Terry model utilizes pairwise comparisons, so only two outcomes, but with soccer there are three outcomes for each team (win, loss, draw) (Firth, 2005). To adjust for this, I averaged the probability of the home team winning or drawing with the probability of the home team only winning when calculating team strength. This could cause my findings to be slightly inaccurate as draws are not worth exactly half of wins. The last major limitation of my model is that it has only been tested on large leagues with 18 or more teams. It is unclear whether my model would accurately measure competitive balance in a smaller league.

Implications

My metric of competitive balance is most useful to those controlling the soccer leagues as it helps them better identify what factors lead to a more balanced league. However, it is clear that the most balanced leagues are not the most revenue inducing. This implicates that competitive balance is not the end all be all in attracting and keeping viewers. This is important as maintaining fan engagement and maximizing revenue potential is the goal of all leagues, and the findings my paper presents can help leagues make informed decisions to reach these goals.

MLS

In Figures 2 - 7, there are a few common trends I can identify. First, the second division in a country's system of professional soccer leagues is generally more competitive than the first division. In addition, I can see how the MLS is much more comparable to those second divisions than it is to other European top flights.

There are a few reasons why the MLS is as competitive as 2nd divisions. For one, the MLS's level of teams may be more comparable to these second division leagues, with players' skill levels being less consistent than those in the top flight allowing for any team to win on a given day. But, the reason I think that the MLS is more competitively balanced is due to its salary cap. This is supported by a paper from Kavi Sachania, which finds that the MLS's flat salary cap has increased the league's competitive balance while keeping wages down at the price of deterring top players from coming to the league (Sachania, 2019). This paper is a little outdated as the MLS now has started to attract bigger and bigger talent with the transfer of Lionel Messi.

Financial Imbalance

The salary cap levels the financial playing field and allows similar level players to play for each team. This characteristic of the MLS is also similar to 2nd divisions as they too have a much more even financial playing field. An example of this can be seen in England where the difference in revenue and average wages between the top team and bottom team in 2023 was almost \$500 million while the difference between the top and bottom teams in the championship was around \$50 million (D.S.B.G., 2023). This financial gap is what most people believe causes the imbalance in top leagues when compared to lower leagues.

Another thing to consider when looking at my results is that the research is only over the set time frame of 2005 to the present, around the time when oil money began coming into European football with Manchester City being bought by Sheikh Mansour in 2008 and Chelsea by Roman Abramovich in 2003 (Basini et al., 2023).

What Imbalance Looks Like

A very competitively balanced season has all the teams in the league around similar points, but what type of season makes for the most competitive imbalance is not so simple. Understanding this will help us determine what sets the MLS and 2nd divisions apart from the other professional soccer leagues analyzed.

By using Julia to filter out the most imbalanced singular seasons, I found that the most competitively imbalanced season comes when either two to three teams are far better than the rest of the league or two to three teams are much worse than the rest of the league. For example, two of the most imbalanced seasons in the English Premier league are 2016-17 and 2018-19 seasons. The 2018-19 season is the Liverpool and Manchester City title race with Manchester City winning on 98 points and Liverpool finishing on 97. The 2016-17 season saw Sunderland and Middlesbrough be almost 10 points worse than the rest of the league while Chelsea and Tottenham were in a two team title race. While this title race is not as extreme, the combination of the gap at the bottom and the top is what makes this season so competitively imbalanced.

Another important aspect of Figure 1 is that leagues like the Dutch, Greek, and Portuguese first divisions have seasons with exceedingly high competitive balance indexes. The reason for this is a little different for each league.

Firstly, the format of the Greek first division is what makes it so imbalanced. Following a 26 game round robin style section where each team plays the other 13 teams twice, the league is split into a top 6 and bottom 8, where the teams in each of these divisions fight for the chance to win the league or not get relegated. This structure artificially makes the league more imbalanced but also more exciting to watch in a different way as it fills the end of the season with competitive high stakes games. This is an illustration of a possible shortcoming of my model as it does not consider structures like these, where exciting games come at the price of competitive

balance. Greek Super League attendance has grown significantly since this new system was implemented with average attendance going from 3,000-5,000 to 6,500-7,500.

When it comes to the Dutch and Portuguese leagues, the imbalance can be attributed to the gap between the top 3-4 teams and the rest of the competition along with some seasons having some teams perform exceedingly poorly. For Portugal in particular, their first division has been dominated by 4 clubs in the modern era: Porto, Sporting Lisbon, Benfica, and Braga. The most recent 2022-23 seasons saw a 20 point gap between the 4th place Sporting Lisbon and 5th place Arouca. On the other hand, the Dutch league has been dominated by PSV, Ajax, and Feyenoord who are almost always well above the rest of the competition. The Dutch league also sees a consistent trend of the bottom team finishing with around 20% of the possible points earned, significantly lower than the rest of the league.

Conclusion

In this study, I set out to address the challenge of quantifying competitive balance in professional men's soccer leagues from 2005 to 2023 through Bradley-Terry model and pairwise while simultaneously evaluating that measure's validity. My research aimed to bridge the gap in existing measures of competitive balance and offer a comprehensive index that accurately captures the dynamics of competitiveness across diverse leagues, thus providing evidence of the comprehensive usefulness of the Bradley-Terry model in such situations.

My findings demonstrated the effectiveness of the Bradley-Terry model in generating a robust metric for evaluating league parity. By utilizing pairwise comparisons and team strengths derived from game outcomes, I developed a novel index that offers insights into the competitive

landscape of soccer leagues worldwide. This measure was shown to be valid through the comparison of its results to those from the likes of Basini et al. and Criado et al.

My analysis of the data revealed compelling implications for the broader community of sports management and economics. Firstly, my study underscores the importance of perceived fairness and league parity in maintaining fan engagement and maximizing revenue potential. However, it also finds that the competitive balance of a league does not triumph all when it comes to maintaining and growing a viewer base. By quantifying competitive balance, leagues can make informed decisions regarding structural interventions aimed at promoting competitiveness and sustainability, leading to an overall better product.

Looking ahead, future research endeavors could build upon my findings to deepen my understanding of competitive balance in sports leagues. Specifically, further exploration of the applicability of the Bradley-Terry model in smaller leagues and non-traditional sports contexts would enhance its utility and generalizability. Additionally, attempting to determine the exact reasons for the differences in competitive balance between leagues, such as league structure, nature of each sport, and wealth inequality, would be a valuable next step. In line with this, longitudinal studies examining the impact of structural interventions on competitive balance would provide valuable insights for league policymakers.

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Appendix

Appendix 1

Functions for calculating competitiveness from mixed models

```
5]: function htwSoccer(x::DataFrame)
    htw = transform!(x, [:FTHG, :FTAG] => ((hg, ag) -> (hg.>ag)) => :HTW)
    subset!(x, :FTHG=> (x->(!ismissing).(x)))
    htw = select(htw, :Date, :HomeTeam, :AwayTeam, :FTHG, :FTAG, :HTW)
    return htw
end

5]: htwSoccer (generic function with 1 method)

6]: function htdwSoccer(x::DataFrame)
    htdw = transform!(x, [:FTHG, :FTAG] => ((hg, ag) -> (hg.>=ag)) => :HTDW)
    subset!(x, :FTHG=> (x->(!ismissing).(x)))
    htdw = select(htdw, :Date, :HomeTeam, :AwayTeam, :FTHG, :FTAG, :HTDW)
    return htdw
end

6]: htdwSoccer (generic function with 1 method)

7]: function compSoccer(x::DataFrame)
    comp = fit(MixedModel, @formula(HTW~1+(1|fulldummy(HomeTeam))+(1|fulldummy(AwayTeam))), htwSoccer(x), Bernoulli())
    vc = VarCorr(comp)
    return (vc.op.HomeTeam.σ[1] + vc.op.AwayTeam.σ[1])/2, coef(comp)[1]
end

7]: compSoccer (generic function with 1 method)

8]: function compSoccerWithDraw(x::DataFrame)
    outWin = fit(MixedModel, @formula(HTW~1+(1|fulldummy(HomeTeam))+(1|fulldummy(AwayTeam))), htwSoccer(x), Bernoulli())
    # check below
    outLoss = fit(MixedModel, @formula(HTDW~1+(1|fulldummy(HomeTeam))+(1|fulldummy(AwayTeam))), htdwSoccer(x), Bernoulli())
    vcWin = VarCorr(outWin)
    vcLoss = VarCorr(outLoss)
    comp = (vcWin.op.HomeTeam.σ[1] + vcWin.op.AwayTeam.σ[1] +
            vcLoss.op.HomeTeam.σ[1] + vcLoss.op.AwayTeam.σ[1])/4
    return comp, (coef(outWin)[1]+coef(outLoss)[1])/2
end

8]: compSoccerWithDraw (generic function with 1 method)
```

Appendix 2

```
function leaguetable(x::DataFrame)
# this function creates a table for soccer league that contains goals for, goals againts, goal difference
# games played, points, 3 home field advantage metrics, and 2 team strength metrics
# select columns we will use
LeagueSet = select(x, :Date, :HomeTeam, :AwayTeam, :FTHG, :FTAG)
# group by home and away teams
teamsH = groupby(LeagueSet, :HomeTeam)
teamsA = groupby(LeagueSet, :AwayTeam)
gfHome = combine(teamsH, :FTHG => sum)
gfHome = rename(gfHome, :HomeTeam => :Team)
gfAway = combine(teamsA, :FTAG => sum)
gfAway = rename(gfAway, :AwayTeam => :Team)
gfTotal = innerjoin(gfHome, gfAway; on=:Team)
gfTotal = rename(gfTotal, :FTHG_sum => :HG, :FTAG_sum => :AG)
gfTotal.TG = gfTotal.HG + gfTotal.AG
gaHome = combine(teamsH, :FTAG => sum)
gaHome = rename(gaHome, :HomeTeam => :Team);
gaAway = combine(teamsA, :FTHG => sum)
gaAway = rename(gaAway, :AwayTeam => :Team);
gdHome = innerjoin(gfHome, gaHome; on=:Team)
gdHome = rename(gdHome, :FTHG_sum => :GFH, :FTAG_sum => :GAH);
gdHome.GDTH = gdHome.GFH - gdHome.GAH;
gdAway = innerjoin(gfAway, gaAway; on=:Team)
gdAway = rename(gdAway, :FTAG_sum => :GFA, :FTHG_sum => :GAA);
gdAway.GDTA = gdAway.GFA - gdAway.GAA;
gdTotal = innerjoin(gdHome, gdAway; on=:Team);
gdTotal.GF = gdTotal.GFH + gdTotal.GFA;
gdTotal.GA = gdTotal.GAH + gdTotal.GAA;
gdTotal.GD = gdTotal.GDTH + gdTotal.GDTA;
LeagueSet.pointsH = pointH.(LeagueSet.FTHG, LeagueSet.FTAG);
LeagueSet.pointsA = pointA.(LeagueSet.FTHG, LeagueSet.FTAG);
ptsHome = combine(teamsH, :pointsH => sum)
ptsHome = rename(ptsHome, :HomeTeam => :Team)
ptsAway = combine(teamsA, :pointsA => sum)
ptsAway = rename(ptsAway, :AwayTeam => :Team)
ptsTotal = innerjoin(ptsHome, ptsAway; on=:Team)
ptsTotal = rename(ptsTotal, :pointsH_sum => :PointsH, :pointsA_sum => :PointsA);
Table = innerjoin(gdTotal, ptsTotal; on=:Team);
Table.Points = Table.PointsH + Table.PointsA;
Table = select(Table, :Team, :GF, :GA, :GD, :Points, :PointsH, :PointsA, :GDTH, :GDTA, :GFH, :GFA)
Table = sort(Table, :Points, rev=true);
Table.Games .= (size(Table,1)-1)*2
```

```
Table.Games .= (size(Table,1)-1)*2
Table = select(Table, :Team, :Games, :GF, :GA, :GD, :Points, :PointsH, :PointsA, :GDTH, :GDTA, :GFH, :GFA)
Table.HFAdvPts = (Table.PointsH - Table.PointsA)./Table.Games;
Table.HFAdvGD = (Table.GDTH - Table.GDTA)./Table.Games
Table.HFAdvGF = (Table.GFH - Table.GFA)./Table.Games
Table.TSPts = (Table.Points)./Table.Games
Table.TSGD = (Table.GD)./Table.Games
Table = select(Table, :Team, :Games, :GF, :GA, :GD, :Points, :HFAdvPts, :HFAdvGD, :HFAdvGF, :TSGD, :TSPts)
return Table
end
```

[3]: leaguetable (generic function with 1 method)